

Education

- 2017 **PhD Statistics**, University of Minnesota Twin Cities, Minneapolis, MN, USA
- 2012 **MS Statistics**, Indian Statistical Institute, Kolkata, India
- 2010 **BS Statistics**, Indian Statistical Institute, Kolkata, India

Experience

- 2026- **Assistant Professor**, Indian Institute of Management Bangalore
Decision Sciences Area
IIMB Young Faculty Research Chair, 2026-28
- 2023-25 **Co-founder, Head of AI**, Vijil
- 2023 **Principal Machine Learning Scientist**, GEICO
- 2022-23 **Applied Scientist II**, Amazon (Twitch)
- 2021-22 **Senior Applied Scientist**, Splunk
- 2018-21 **Senior Inventive Scientist**, AT&T Labs Research
- 2017-18 **Postdoctoral Researcher**, University of Florida
- 2016 **Research Intern**, IBM Research

Research Interests

Trustworthy AI/ML, AI Security, AI Alignment, Statistical machine learning.

Publications

Theory and Methods

- 2025 H. Raj, V. Gupta, D. Rosati, **S. Majumdar**. Improving Consistency in Large Language Models through Chain of Guidance. *Transactions on Machine Learning Research*.
- 2024 D. Rosati, J. Wehner, K. Williams, L. Bartoszcze, D. Atanasov, R. Gonzales, **S. Majumdar**, C. Maple, H. Sajjad, F. Rudzicz. Representation Noising: A Defence Mechanism Against Harmful Finetuning. *Neural Information Processing Systems (NeurIPS)*, 37, 12636-12676.
- 2023 F.T. Brito, V.A.E. Farias, C. Flynn, J.C. Machado, **S. Majumdar**, D. Srivastava. Global and local differentially private release of count-weighted graphs. *Proceedings of the ACM on Management of Data (SIGMOD)*, 1 (2), 1-25.
- 2023 R. Rustamov, **S. Majumdar**. Intrinsic sliced wasserstein distances for comparing collections of probability distributions on manifolds and graphs. *International Conference on Machine Learning (ICML)*, 29388-29415.
- 2023 V.A.E. Farias, F.T. Brito, C. Flynn, J.C. Machado, **S. Majumdar**, D. Srivastava. Local Dampening: Differential Privacy for Non-numeric Queries via Local Sensitivity. *The VLDB Journal*, 32, 1191-1214.

- 2022 **S. Majumdar**, S. Chatterjee. Feature selection using e-values. *International Conference on Machine Learning (ICML)*, 14753-14773.
- 2022 **S. Majumdar**, G. Michailidis. Joint estimation and inference for data integration problems based on multiple multi-layered gaussian graphical models. *Journal of Machine Learning Research*, 23, 1-53.
- 2022 **S. Majumdar**, S. Chatterjee. On weighted multivariate sign functions. *Journal of Multivariate Analysis*, 105013.
- 2020 A. Ghosh, **S. Majumdar**. Ultrahigh-dimensional Robust and Efficient Sparse Regression using Non-Concave Penalized Density Power Divergence. *IEEE Transactions on Information Theory*, 66 (12), 7812-7827.
- 2018 **S. Majumdar**, S. Chatterjee. Non-convex penalized multitask regression using data depth-based penalties. *Stat*, 7, e174.

Applications

- 2025 B. Barazandeh, **S. Majumdar**, O. Rajyaguru, G. Michailidis. Localized LoRA: A Structured Low-Rank Approximation for Efficient Fine-Tuning. *International Conference on Machine Learning and Applications (ICMLA)*, accepted.
- 2025 R. Clancy, Q. Zhu, **S. Majumdar**. Exploring AI Ethics in Global Contexts: A Culturally Responsive, Psychologically Realist Approach. *AI and Ethics*, 5, 6329-6338.
- 2025 **S. Majumdar**, B. Pendleton, A. Gupta. Red Teaming AI Red Teaming. *Conference on Applied Machine Learning in Information Security (CAMLIS)*, PMLR 299, 66-86.
- 2024 M.A. Ayub, **S. Majumdar**. Embedding-based classifiers can detect prompt injection attacks. *Conference on Applied Machine Learning in Information Security (CAMLIS)*.
- 2023 **S. Majumdar**, S. Basu, M. McGue, S. Chatterjee. Simultaneous selection of multiple important single nucleotide polymorphisms in familial genome wide association studies data. *Scientific Reports*, 13 (1), 8476.
- 2022 G. Subramaniam, **S. Majumdar**. Network Security Modelling with Distributional Data. *Conference on Applied Machine Learning in Information Security (CAMLIS)*.
- 2021 N. Derzsy, **S. Majumdar**, R. Malik. An Interpretable Graph-based Mapping of Trustworthy Machine Learning Research. *International Conference on Complex Networks (CompleNet)*.
- 2020 S.P. Fookolaee, S. Karkhah, M. Saadi, **S. Majumdar**, A. Karkhah. Novel computational approaches to developing potential STAT4 silencing siRNAs for immunomodulation of atherosclerosis. *Current Computer Aided Drug Design*, 16 (5), 599-604.
- 2019 S.C. Basak, **S. Majumdar**, and others. Computer-Assisted and Data Driven Approaches for Surveillance, Drug Discovery, and Vaccine Design for the Zika Virus. *Pharmaceuticals*, 12, 157.
- 2019 **S. Majumdar**, S.C. Basak, and others. Finding needles in a haystack: determining key molecular descriptors associated with the blood-brain barrier entry of chemical compounds using machine learning. *Molecular Informatics*, 38, 1800164.
- 2019 B. Han, **S. Majumdar**, and others. Confronting data sparsity to identify potential sources of Zika virus spillover infection among primates. *Epidemics*, 27, 59-65.
- 2018 **S. Majumdar**, S.C. Basak. Beware of external validation! – A Comparative Study of Several Validation Techniques used in QSAR Modelling. *Current Computer Aided Drug Design*, 14, 284-291.

- 2018 **S. Majumdar**, S.C. Basak, and others. Mathematical structural descriptors and mutagenicity assessment: A study with congeneric and diverse data sets. *SAR and QSAR in Environmental Research*, 29, 579–590.
- 2016 **S. Majumdar**, S.C. Basak. Exploring intrinsic dimensionality of chemical spaces for robust QSAR model development: A comparison of several statistical approaches. *Current Computer Aided Drug Design*, 12, 294–301.
- 2015 S.C. Basak, **S. Majumdar**. Prediction of Mutagenicity of Chemicals from Their Calculated Molecular Descriptors: A Case Study with Structurally Homogeneous versus Diverse Datasets. *Current Computer Aided Drug Design*, 11, 117–123.
- 2015 E. Potash, J. Brew, A. Loewi, **S. Majumdar**, A. Reece, J. Walsh, E. Rozier, E. Jorgenson, R. Mansour, and R. Ghani. Predictive Modeling for Public Health: Preventing Childhood Lead Poisoning. *21st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2039–2047.
- 2013 **S. Majumdar**, S.C. Basak, G.D. Grunwald. Adapting Interrelated Two-Way Clustering Method for Quantitative Structure-Activity Relationship (QSAR) Modeling of Mutagenicity/Non-Mutagenicity of a Diverse Set of Chemicals. *Current Computer Aided Drug Design*, 9, 463–471.

Preprints

- 2026 D. Rosati, X. Zeng, H. Huang, S. Dionicio, **S. Majumdar**, F. Rudzicz, H. Sajjad. Limits of Convergence-Rate Control for Open-Weight Safety. *arXiv:2602.18868*.
- 2024 L. Derczynski, E. Galinkin, J. Martin, **S. Majumdar**, N. Inie. garak: A Framework for Security Probing Large Language Models. *arXiv:2406.11036*.
- 2021 **S. Majumdar**, S. Chatterjee. Generalized Multivariate Signs for Nonparametric Hypothesis Testing in High Dimensions. *arXiv:2107.01103*.

Books

- 2023 Y. Pruksachatkun, M. Mcateer, **S. Majumdar**. *Practicing Trustworthy Machine Learning: Consistent, Transparent, and Fair AI Pipelines*. O'Reilly Media.

Book Chapters

- 2024 **S. Majumdar**. *Standards for LLM Security*. In: Large Language Models in Cybersecurity, Springer, 225–231.
- 2024 **S. Majumdar**, T. Vogelslang. *Towards Safe LLMs Integration*. In: Large Language Models in Cybersecurity, Springer, 243–247.
- 2019 **S. Majumdar**. *Data-driven Strategies to Model and Mitigate the Threat of Zika*. In: Zika virus: Basic biology, surveillance, vaccine design and anti-Zika drug discovery: Computer-assisted strategies to combat the menace, Nova Science Publishers, Inc., 129–152.
- 2015 S.C. Basak, **S. Majumdar**. *Current Landscape of Hierarchical QSAR Modeling and its Applications: Some Comments on the Importance of Mathematical Descriptors as well as Rigorous Statistical Methods of Model Building and Validation*. In: Advances in Mathematical Chemistry and Applications: Vol. 1, Elsevier and Bentham e-Books, 251–281.
- 2015 U. Mukherjee, **S. Majumdar**, S. Chatterjee, *Fast and Robust Supervised Learning in High Dimensions Using the Geometry of the Data*. In: Advances in Data Mining: Applications and Theoretical Aspects, ser. Lecture Notes in Computer Science, 9165, 109–123.

Refereed Workshops

- 2025 D. Rosati, S. Dionicio, X. Zeng, **S. Majumdar**, F. Rudzicz, H. Sajjad. Locking Open Weight Models with Spectral Deformation. *ICML 2025 Workshop on Technical AI Governance*.
- 2025 J. Novikova, C. Anderson, B. Blili-Hamelin, D. Rosati, **S. Majumdar**. Consistency in Language Models: Current Landscape, Challenges, and Future Directions. *ICML 2025 Workshop on Reliable and Responsible Foundation Models*.
- 2025 D. Rosati, G. Edkins, H. Raj, D. Atanasov, **S. Majumdar**, J. Rajendran, F. Rudzicz, H. Sajjad. Mitigating Unsafe Feedback with Learning Constraints, *AAAI 2025 Workshop on Artificial Intelligence for Cyber Security*.
- 2022 H. Raj, D. Rosati, **S. Majumdar**. Measuring Reliability of Large Language Models through Semantic Consistency. *NeurIPS 2022 ML Safety Workshop (Best paper award)*.
- 2022 C. Flynn, A. Guha, **S. Majumdar**, D. Srivastava, Z. Zhou. Towards Algorithmic Fairness in Space-Time: Filling in Black Holes. *NeurIPS 2022 Workshop on Trustworthy and Socially Responsible Machine Learning*.
- 2022 **S. Majumdar**, C. Flynn, R. Mitra. Detecting bias in the presence of spatial autocorrelation. *NeurIPS 2021 Algorithmic Fairness through the Lens of Causality and Robustness workshop*.
- 2021 C. Last, P. Pramanik, N. Saini, A.S. Majety, D.-H. Kim, M. García-Herranz, **S. Majumdar**. Towards an Open Global Air Quality Monitoring Platform to Assess Children's Exposure to Air Pollutants in the Light of COVID-19 Lockdowns. *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*.

Patents

Please find a list of 20+ filed and granted patents here.

Talks

Keynotes

- June 2023 CVPR Workshop on Fair, Data Efficient and Trusted Computer Vision, Vancouver, Canada
- Sep 2022 8th Indo-US Workshop on Mathematical Chemistry, virtual
- Aug 2022 Faculty Development Programme, Saranathan College of Engineering, Trichy, India
- July 2022 NAACL Workshop on Trustworthy Natural Language Processing, Seattle, WA

Invited

- Aug 2026 CMStatistics EcoSta 2026 conference, Kyoto, Japan
- June 2026 International Symposia on Nonparametric Statistics (ISNPS), Thessaloniki, Greece
- June 2025 International Indian Statistical Association (IISA) Conference, Lincoln, NE
- June 2025 Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai, India
- May 2025 AI Ethics Education Workshop, University of Alabama, Tuscaloosa, AL
- Feb 2025 University of Southern California Marshall School of Business, Los Angeles, CA
- Jan 2025 International Conference on Data Management, Analytics & Innovation, Kolkata, India
- Apr 2024 LinkedIn, Bellevue, WA
- Aug 2023 O'Reilly Expert Webinar, virtual
- Nov 2022 Open Data Science Conference West, San Francisco, CA

Apr 2022 University of Washington RAISE lab, Seattle, WA
 Dec 2020 Data Science Salon, virtual
 Nov 2020 (Lecture series) Indian Institute of Technology Kanpur, Department of Mathematics and Statistics, India
 Mar 2019 Women in Machine Learning and Data Science meetup, New York, NY
 May 2018 IISA Conference, Gainesville, FL
 May 2018 Savvysherpa, Inc., Minneapolis, MN
 Dec 2017 IISA Conference, Hyderabad, India
 Dec 2017 Indian Statistical Institute, Kolkata, India
 Aug 2016 (Student paper) IISA Conference, Corvallis, OR

Panels

Nov 2025 AI Safety Dublin, virtual (fireside chat)
 June 2025 International Indian Statistical Association (IISA) Conference, Lincoln, NE
 May 2025 AI Ethics Education Workshop, University of Alabama, Tuscaloosa, AL
 Jan 2025 National Association of Software and Service Companies (NASSCOM), Kolkata, India
 Dec 2024 International Indian Statistical Association (IISA) Conference, Kochi, India
 Sep 2024 OctoAI Builders Roundtable: Secure GenAI for Enterprises, virtual
 Oct 2022 ML: Integrity Conference, virtual
 Feb 2020 National Institute of Statistical Sciences (NISS) Industry Career Fair, virtual

Awards

2026–28 Young Faculty Research Chair, Decision Sciences, IIM Bangalore
 2024 IISA Early Career Award in Statistics and Data Sciences
 2016–17 University of Minnesota Interdisciplinary Doctoral Fellowship
 2016–17 University of Minnesota School of Statistics Martin Award
 2016 Best Student Paper, IISA conference, Corvallis, OR
 2016 IISA Conference best student paper award
 2015 5th International Workshop on Climate Informatics travel award
 2014–16 University of Minnesota School of Statistics travel award
 2012 Debesh-Kamal Scholarship, Ramakrishna Mission Institute of Culture, Kolkata, India
 2008–12 KVPY national fellowship, Department of Science and Technology, Govt. of India
 2005–08 National scholar, National Council of Educational Research and Training, Govt. of India

Advising and Mentorship

2026– Muberra Ozmen, independent researcher / research advisor
 2022– Harsh Raj, MS student at Northeastern Univ / internship mentor, research advisor
 2022– Domenic Rosati, PhD student at Dalhousie University / research advisor
 2024–25 Aditya Karan, PhD student at UIUC / internship mentor, research advisor
 2024 Md. Ahsan Ayub, Postdoc at Vanderbilt University / research advisor
 2019–21 Felipe Brito, Universidade Federal do Ceara, Brazil / research advisor

2019–21 Victor Farias, Universidade Federal do Ceara, Brazil / research advisor
2020 Christina Last, MS at Massachusetts Institute of Technology / internship mentor
2020 Prithviraj Pramanik, AQAI / internship mentor

Teaching

School of Statistics, Univ. of Minnesota (as Teaching Assistant)

- Fall 2014 STAT 8051 - Advanced Regression Techniques
- Spring 2014 STAT 3022 - Data Analysis
- Fall 2013 STAT 5021 - Statistical Analysis
- STAT 5031 - Statistical Methods for Quality Improvement
- Spring 2013 STAT 5303 - Designing Experiments
- STAT 5401 - Applied Multivariate Methods
- Fall 2012 STAT 3011 - Introduction to Statistical Analysis

Service

Reviewing

- Journals Sankhya B (Associate Editor), IEEE Transactions on Information Theory, Statistica Sinica, Scientific Reports, Biometrics, R Journal, Applied Computing and Informatics, Current Computer-Aided Drug Design, Australasian Medical Journal
- Conferences AAAI, AISTATS, CAMLIS, ICML, IAAI, NeurIPS, PAKDD
- Workshops TrustNLP at NAACL, ML-RSA at NeuRIPS, CHI Extended Abstracts

Organizing

- 2025 Program Committee member, IISA Conference
- 2024–25 Co-secretary and IT Committee Chair, IISA
- 2021 Organizing team, Trustworthy ML Symposium
- 2017–18 Session Chair, IISA Conferences